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Exercise Sheet 04

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Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. **First-Order Predicate Logic**

   (a) Translate the following sentences into predicate logic: [1P]

   a) Every problem has a solution.
   b) All humans are mortal.
   c) Every problem has a solution that no person understands.
   d) If every problem has a solution then no person exists that understands every problem.

   (b) Show that $\neg(\forall X(p(X) \Rightarrow q(X))) \equiv \exists X(p(X) \wedge \neg q(X))$. [1P]

   (c) Negate the following statement [1P]

   $$\exists A(x(A) \wedge \neg y(20, A) \Rightarrow \forall B(z(A, B)))$$

   (d) Evaluate the truth value of the first-order predicate logic formula α [2P]

   $$\forall x[\neg Q(x, y) \wedge \neg P(f(x)) \Rightarrow R(f(x), x)]$$

   for the following Model $\mathcal{M}$:

   - Universe $U = \{0, 1, 2, 3, 4\}$
   - Predicate Semantics:
     - $P = \{0, 3, 4\}$
     - $Q = \{(0, 1), (0, 2), (3, 1)\}$
     - $R = \{(0, 0), (4, 2), (4, 3), (1, 3)\}$
   - Function semantics: $f(x) = 2x \mod 5$
   - free variable assignments: $y \leftarrow 1$

   (e) Transform the first order predicate formula [2P]

   $$\forall X[\forall Y(P(X, Y) \vee Q(Y)) \Rightarrow Q(X)]$$

   into an equisatisfiable formula in CNF

2. **Logical Reasoning and Inference**

   (a) Show that $\neg(p \wedge q) \vee (p \vee q)$ is a tautology. [1P]

   (b) Prove that superman doesn't exist based on the following knowledgebase: [2P]

   - If Superman were able (A) and willing (W) to prevent(P) evil, he would do so.
   - If Superman were unable to prevent evil, he would be incapable (I)
   - If Superman were unwilling to prevent evil, he would be malevolent (M).
   - Superman does not prevent evil.
   - If Superman exists (E), he is neither incapable nor malevolent.